

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech (Chemical Engineering)	Indian Institute of Technology Kanpur	8.3/10
2021	CBSE (XII)	Delhi Public School, Azaad Nagar, Kanpur	98.0 %
2019	CBSE (X)	Delhi Public School, Azaad Nagar, Kanpur	97.4 %

Scholastic Achievements	
- Secured All India Rank 3267 in Joint Entrance Examination Advanced 2021 among the 2 Lakh shortlisted candidates - Secured All India Rank 2267 in Joint Entrance Examination Mains 2021 among the 22 Lakh registered candidates - Cleared Pre-Regional Mathematics Olympiad 2019 and qualified for RMO 2019 conducted by Homi Bhabha Centre for Science Education	

Key Projects

Algorithmic Graph Theory - Association of Computing Activities, IIT Kanpur (May'23- Jul'23)

Objective	To explore graph theory from an algorithmic perspective and utilise it to effectively solve real-world complex problems
Summary	- Implemented BFS & DFS for traversal, and Dijkstra, Floyd-Warshall & Bellman-Ford Algorithms for shortest path problems - Applied Prim's & Kruskal's Algorithms for MSTs and Kosaraju & Tarjan's Algorithms for Strongly Connected Components - Solved complex graph coloring and network flow problems and selected problems of CSES problemset and Project Euler - Learned to solve Rubik's Cube and Merlin's Machine by applying principles derived from Graph Theory and Group theory

Build your own Computer Vision Library - Stamatics, IIT Kanpur (May'23- Jul'23)

Objective	To develop a comprehensive library in C programming language, encompassing functions for various image processing tasks
Summary	- Developed functions for grayscaleing images using Luma technique and inter-converting between RGB and HSV colourspaces - Executed image-resizing using nearest neighbour & bilinear interpolation and generated function for convolving an image - Created various filters including highpass filter, sharpen filter, emboss filter and used gaussian filter to create hybrid images - Implemented edge detection using sobel filter and studied Difference-of-Gaussian and Canny Edge Detection techniques

Introduction to Data Science - Simutech Winter Project, Chemineers Society, IIT Kanpur (Dec'22- Jan'23)

Objective	To study theorotical concepts and practical applications of Data Science, and implement standard Machine Learning models
Approach	- Developed a linear regression model and a polynomial regression model from scratch using only Numpy & Pandas libraries - Implemented a linear regression model using Sklearn library and studied Logistic Regression & Deep Learning fundamentals - Created a 3- layer Artificial Neural Network from scratch, utilising Backpropagation algorithm and Relu activation function
Results	Implemented the ANN to predict the net hourly electrical energy output of a power plant, and obtained an r2_score of 0.83

MATLAB Computational Project - ESO208 Course Project, Prof. Sachchida Nand Tripathi, IIT Kanpur (Aug'22- Nov'22)

Objective	To develop Matlab codes for executing numerical methods used to solve complex mathematical problems and generate plots
Summary	- Implemented Regular-Falsi, Newton-Raphson, Bairstow and Muller methods for finding roots of arbitrary-degree polynomial - Applied Cholesky, Doolittle and Crout methods for LU decompositon and QR method for finding eigenvalues of a matrix - Used Romberg Integration & Gauss-Legendre Quadrature for numerical integration and Runge-Kutta methods for IVPs - Employed Lagrange polynomials & natural cubic splines for curve interpolation and least-squares regression for data fitting

Automated Chessboard - Electronics Club, Science and Technology Council, IIT Kanpur (May'22- Aug'22)

Objective	To build a one-player physical chessboard with automated movement of chess pieces & computer-driven optimal gameplay
Approach	- Adopted Micro-Max chess program for the optimal gameplay, which is based on MiniMax Algorithm & Alpha-Beta pruning - Developed a complex circuit utilizing Arduino, featuring an electromagnetic moving trolley operated on the core XY system, stepper motors controlled by A4988 stepper motor drive, and reed switches to accurately track the position of chess pieces

Technical Skills

- Programming Languages & Libraries** :- C++, C, Python, HTML, CSS, Numpy, Pandas, Sklearn, Matplotlib
- Software and Utilities** :- Git, GitHub, Matlab, Jupyter Notebook, Arduino, L^AT_EX, Aspen, Canva, Rush, Illustrator

Positions of Responsibility

Manager, Career Development Wing - Academics and Career Council, IIT Kanpur (May'23-Present)

Leadership	Spearheaded a team of 18 secretaries, selected from a pool of 150+ potential candidates through a rigorous recruitment process
Operations	- Organised Internship Preparation Sessions for various major profiles, ensuring comprehensive guidanance for participants - Collaborated with Programming Paathshala for providing a DSA course, including mock test series and mock interviews - Launched Internship Mentorship Program, matching interested students with mentors specializing in their targeted profiles
Impact	Significantly elevated the overall preparation level of over 1000 students appearing for the campus internship drive 2023-24

Company Coordinator - Students' Placement Office, IIT Kanpur (June'22-May'23)

Leadership	Collaborated within a 4-tier team to conduct the Internship Drive for 1000+ students and Placement Drive for 1300+ students
Operations	- Acted as the primary contact for companies like Accenture-Japan and Eaton, ensuring a smooth and efficient interview process - Coordinated with company officials to facilitate pre-internship & pre-placement tests and manage talks & Group Discussions

Senior Executive - Media & Publicity, Techkriti'23, IIT Kanpur (Aug'22-Mar'23)

Leadership	Formed and coordinated a team of 15+ junior executives to ensure the smooth conduction of the activites of the festival
Operations	- Collaborated with professional photographers and social media influencers for recording the key moments of the festival - Associated with different advertising companies and popular tech channels for sponsorship and for the publicity of the festival

Extra-Curricular Activities

Social	- Mentored and aided 6 freshman students to ensure their smooth transition to campus life as Student Guide (2022) - Actively volunteered in Health Camp, organised for the campus workers by Techkriti in collaboration with Amar Ujala (2023)
Cultural	- Successfully delivered a stand-up comedy routine captivating an audience of over 1300 individuals in Orientation'22 (2022) - Participated in Dank Knight, a thrilling event of Thomso'22, the esteemed cultural festival of IIT Roorkee (2022)
Others	Designed promotional posters for events of different councils, which were published through their official handles (2022)